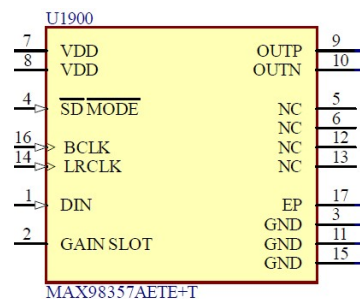


CARRIERTRONIC

Full Service Embedded Provider

How to Use MAX98357A with TDX Carrier Boards and Modules



Status: October.2024, v1.0



Technical changes

carriertronic GmbH reserves the right to change and adapt the information, designs and technical data contained in this documentation without prior notice.

Revision Historie

The following versions of these instructions for use have already been published:

Date	HW-Version	Doc. Version	Changes
October.2024	-	V1.0	Initinal release



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2 Introduction

This guide explains how to connect the **Verdin Pico-ITX Dual Ethernet Carrier Board** from **carriertronic GmbH** to either the **MAX98357A I2S Class-D Amplifier Breakout Board** from **Adafruit** or a custom-built **Verdin Pico-ITX MAX98357A Extension Board** for audio output via the I2S interface. The connection methods for both the Breakout Board and the Extension Board are covered, allowing for flexible integration depending on your setup.

We also reference our **Altium Design Resources**, enabling customers to integrate this setup into their own layout or carrier board extension alongside the **Verdin Pico-ITX Dual Ethernet Carrier Board**.

Furthermore, this How-To will cover the hardware setup as well as additional steps for implementation in Linux, including **Device Tree Overlays** and **Volume Control**.

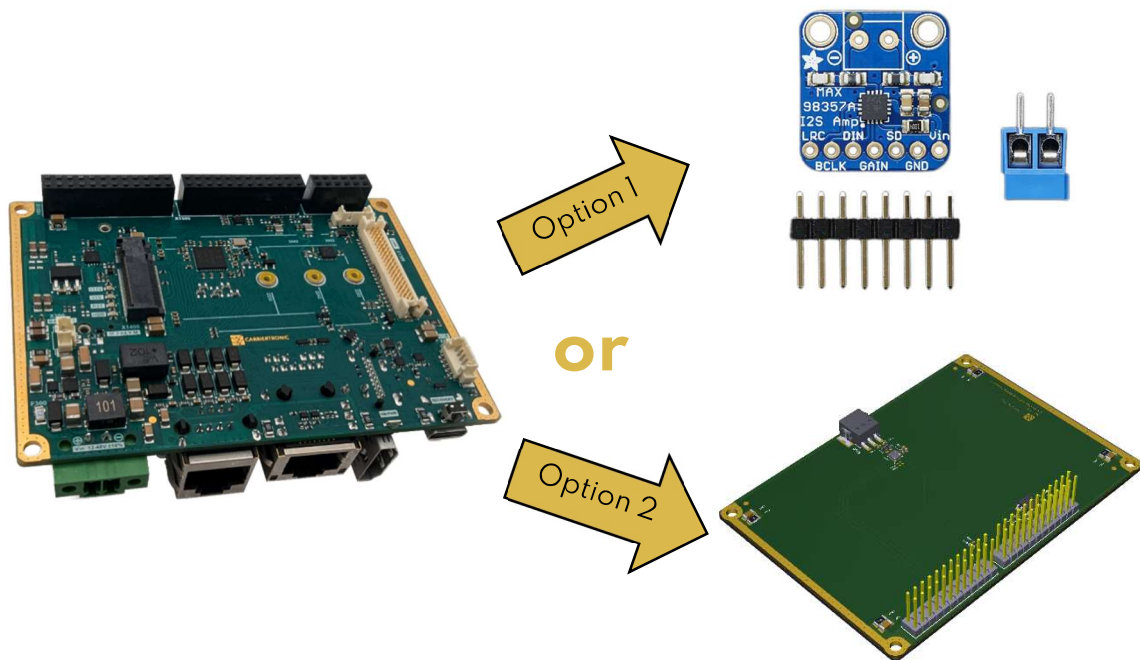
Prerequisites:

Option 1:

- Verdin Pico-ITX Dual Ethernet Carrier Board
- MAX98357A I2S Breakout Board from Adafruit
- Jumper wires for connections
- A suitable power supply for the Carrier Board (e.g. 12 – 48V DC) or PoE

Option 2:

- Verdin Pico-ITX Dual Ethernet Carrier Board
- Verdin Pico-ITX MAX98357A Extension Board (custom-built)
- 4x M3 17mm SSTII Steel Spacer Stud - Metric, internal/internal
- 8x M3x6 screws (Pan head)
- A suitable power supply for the Carrier Board (e.g. 12 – 48V DC) or PoE

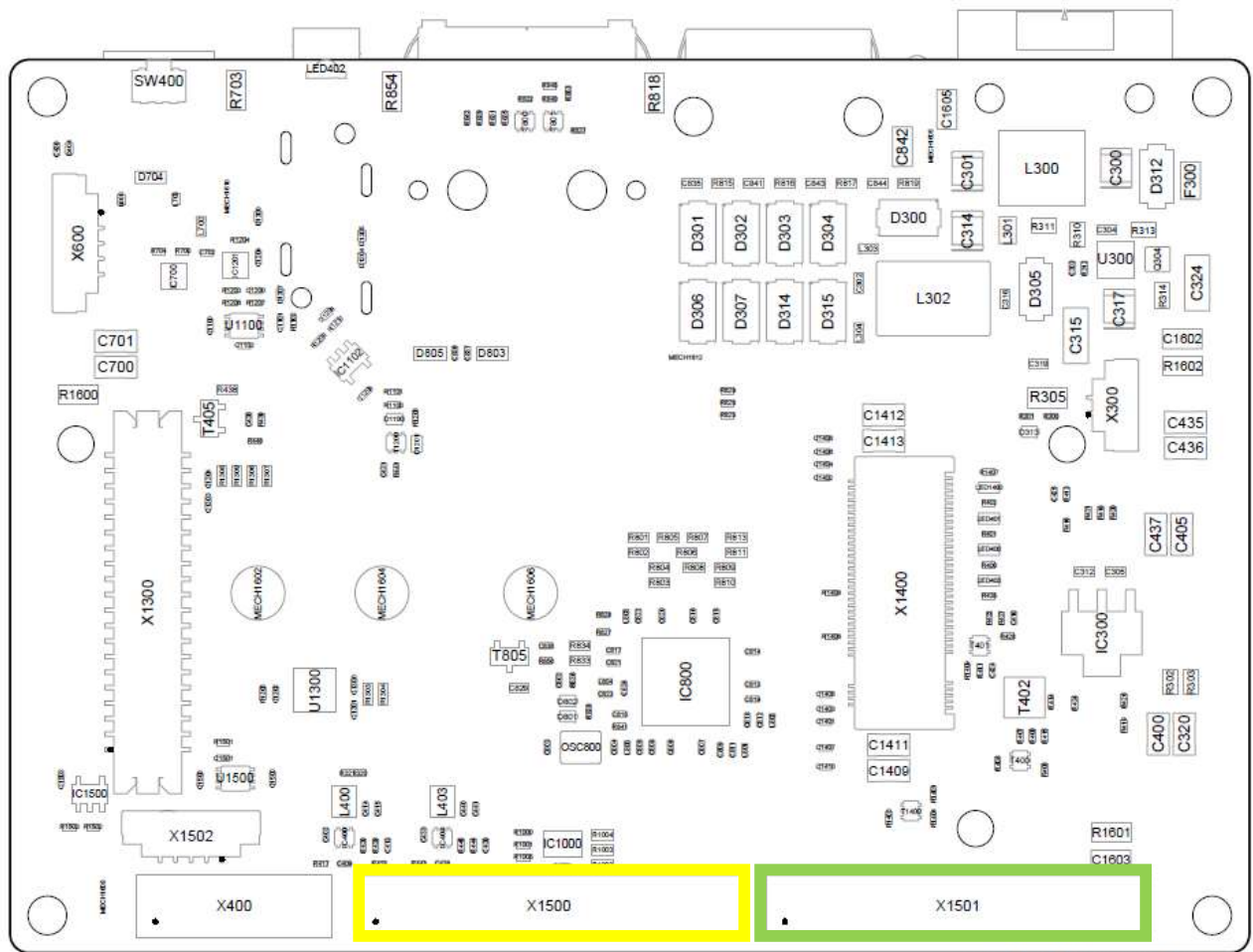




3 Hardware Setup - Option 1

3.1 Step 1: Identify the I2S Pins on the Carrier Board

Ref.	Bottom Side Interfaces
X400	Secondary Extension Header (Debug UART / System Control Signals)
X1502	Capacitive Touch-/Ambient Light Sensor
X1500	Primary Extension Header (Supplies / 2x I2C / 1x PWM / 4x GPIO)
X1501	Primary Extension Header (1x SPI / 2x CAN / 1x I2S / 2x UART)
X1400	1x PCIe M.2 Key-M*
X1300	LVDS*
X300	RTC Backup Power
X600	USB 2.0 Host



View from Bottom side (Scale 5:2)



HowTo: Use MAX98357A with TDX Carrier Boards and Modules

Connector Typ: 2x12 Pin Header Female, 2.54mm Pitch

Pin assignments for X1501

Pin	Connection	Type	SODIMM	Voltage	Description
1	SPI_1_CLK	O	196	+1.8V	SPI Serial Clock
2	SPI_1_MISO	I	198	+1.8V	SPI Master Input, Slave Output
3	SPI_1_MOSI	O	200	+1.8V	SPI Master Output, Slave Input
4	SPI_1_CS	O	202	+1.8V	SPI Chip Select
5	GND	PWR			
6	CAN_1_TX	O	20	+1.8V	CAN port 1 transmit pin
7	CAN_1_RX	I	22	+1.8V	CAN port 1 receive pin
8	CAN_2_TX	O	24	+1.8V	CAN port 2 transmit pin
9	CAN_2_RX	I	26	+1.8V	CAN port 2 receive pin
10	GND	PWR			
11	I2S_1_BCLK	O	30	+1.8V	Serial audio bit clock
12	I2S_1_SYNC	O	32	+1.8V	Synchronization/ field select/ left-right channel select
13	I2S_1_D_OUT	O	34	+1.8V	Serial audio output data
14	I2S_1_D_IN	I	36	+1.8V	Serial audio input data
15	I2S_1_MCLK	O	38	+1.8V	Serial audio master clock
16	GND	PWR			
17	UART_1_RXD	I	129	+1.8V	UART1 Receive Data
18	UART_1_TXD	O	131	+1.8V	UART1 Transmit Data
19	UART_1_RTS	I	133	+1.8V	UART1 Request to Send (RTS)
20	UART_1_CTS	O	135	+1.8V	UART1 Clear to Send (CTS)
21	UART_2_RXD	I	137	+1.8V	UART2 Receive Data
22	UART_2_TXD	O	139	+1.8V	UART2 Transmit Data
23	UART_2_RTS	I	141	+1.8V	UART2 Request to Send (RTS)
24	UART_2_CTS	O	143	+1.8V	UART2 Clear to Send (CTS)

Connector Typ: 2x12 Pin Header Female, 2.54mm Pitch

The pin assignments for X1500

Pin	Connection	Type	SODIMM	Voltage	Description
1	+V5_SW	PWR		+5V	Power Supply Output
2	+V3.3_SW	PWR		+3.3V	Power Supply Output
3	+V1.8_SW	PWR		+1.8V	Power Supply Output
4	GND	GND			
5	Reserve				
6	Reserve				
7	Reserve				
8	Reserve				
9	GND	PWR			
10	I2C_1_SDA	IO	12	+1.8V	Generic I ² C Data
11	I2C_1_SCL	IO	14	+1.8V	Generic I ² C Data
12	I2C_2_DSI_SDA	IO	55	+1.8V	MIPI DSI I ² C Data
13	I2C_2_DSI_SCL	IO	53	+1.8V	MIPI DSI I ² C Data
14	Reserve				
15	Reserve				
16	I2C_4_CSI_SDA	IO	95	+1.8V	MIPI CSI I ² C Data
17	I2C_4_CSI_SCL	IO	93	+1.8V	MIPI CSI I ² C Data
18	PWMI	O	15	+1.8V	General-purpose PWMI
19	Reserve				
20	GND	PWR			
21	GPIO_1	IO	206	+1.8V	General-purpose IO
22	GPIO_2	IO	208	+1.8V	General-purpose IO
23	GPIO_3	IO	210	+1.8V	General-purpose IO
24	GPIO_4	IO	212	+1.8V	General-purpose IO



3.2 Step 2: Identify the Connection on the MAX98357A Breakout Board

The MAX98357A Breakout Board has the following relevant pins:

Pin	Connection	Type	Voltage	Description
1	VDD	PWR	+5V	Positiv Supply for Breakout Board
2	GND	PWR	0V	Negativ Supply for Breakout Board
3	SD_MODE	O	+1.8V	Shutdown control
4	GAIN	I	+1.8V	Gain Control
5	DIN	I	+1.8V	I2S Data In
6	BCLK	I	+1.8V	I2S Bit Clock
7	LRCLK	I	+1.8V	I2S Left/Right Clock

3.3 Step 3: Connect the I2S Lines

I2S Bit Clock (BCLK)

- Connect **Pin 11 (I2S_1_BCLK)** from the Carrier Board (X1501) to the **BCLK** pin 6 on the MAX98357A Breakout Board.

I2S Left/Right Clock (LRC)

- Connect **Pin 12 (I2S_1_SYNC)** from the Carrier Board (X1501) to the **LRC** pin 7 on the MAX98357A Breakout Board.

I2S Data (DIN)

- Connect **Pin 13 (I2S_1_D_OUT)** from the Carrier Board (X1501) to the **DIN** pin 5 on the MAX98357A Breakout Board.

I2S SD_Mode

- Connect **Pin 14 (I2S_1_D_IN)** from the Carrier Board (X1501) to the **SD_MODE** pin 3 on the MAX98357A Breakout Board.

3.4 Step 4: Power and Ground Connections

Power (VDD)

- Connect the **VDD** pin on the MAX98357A Breakout Board to a 3.3V or 5V power source. The Verdin Pico-ITX Carrier Board offers a 3.3V Power supply on Pin 2 (**+V5_SW**) or 5V power supply on Pin 1 (**+V5_SW**) from the Primary Extension Header X1500

Ground (GND)

- Connect the **GND** pin on the MAX98357A Breakout Board to any ground pin on the Carrier Board, such as Pin 4 or Pin 9 (**GND**) on the X1500 header



3.5 Step 5: Additional Control (Optional)

GAIN Selection:

- You can adjust the gain of the MAX98357A by controlling the Gain Pin.

GAIN_SLOT	I2S/LJ GAIN (dB)
Connect to GND through 100k $\pm$ 5% resistor	15
Connect to GND	12
Unconnected	9
Connect to VDD	6
Connect to VDD through 100k $\pm$ 5% resistor	3

Once the physical connections are made, your MAX98357A Breakout Board should be ready to receive audio data from the Verdin Pico-ITX Dual Ethernet Carrier Board via the I2S interface.



4 Hardware Setup - Option 2

4.1 Assembly instructions

- Place the **Verdin Pico-ITX Dual Ethernet Carrier Board** on top of the **Verdin Pico-ITX MAX98357A Extension Board**.
- Align the **X1500** and **X1501 Primary Extension Headers** on both boards and connect them together.
- Use the **4x M3 17mm SSTII Steel Spacer Studs** as spacers between the two boards for structural support.
- Secure the boards using **8x M3x6 screws (pan head)** to fasten the spacer studs to the boards.

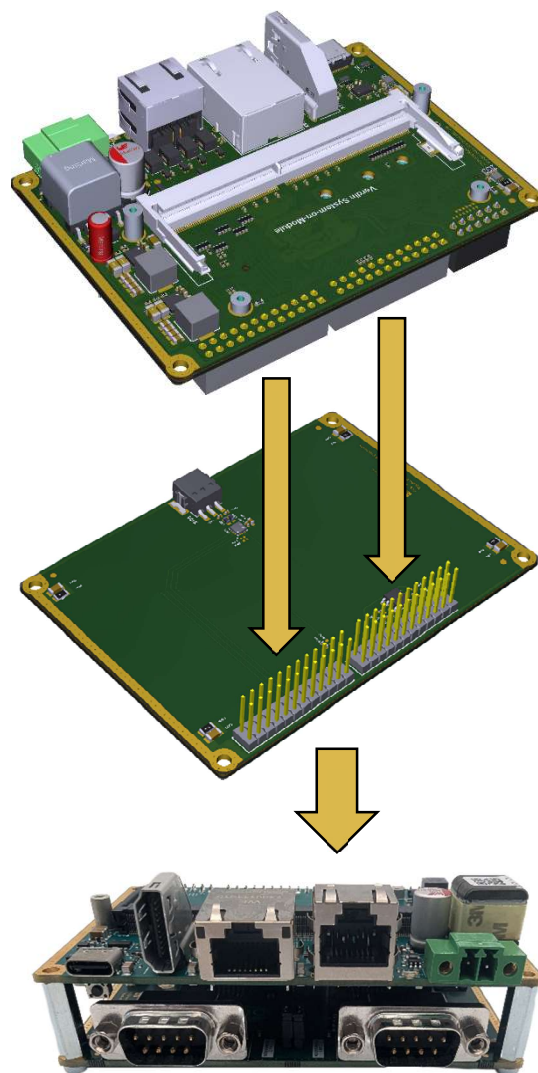


Illustration is only exemplary and shows the connection with our Industrial Extension



5 Device Tree Overlay

Copy and add the DeviceTree Overlay to the System.
This Overlay is tested for Kernel Version 5.15.129-6.4.0

verdin-imx8mp_max98357a_overlay.dtbo

Information on how to modify and compile your own DeviceTree Overlays can be found on the Toradex Developer page under the following link:

<https://developer.toradex.com/software/linux-resources/device-tree/first-steps-with-device-tree-overlays/>



5.1 Devicetree Overlay dtso

Device Tree overlay *verdin-imx8mp_max98357a_overlay.dtso*

```
dts-v1/;
/plugin/;
#include <dt-bindings/clock/imx8mp-clock.h>
#include <dt-bindings/gpio/gpio.h>
/{
    compatible = "toradex,verdin-imx8mp";
};
&{/}{
    // Codec
    codec_ext: max98357a {
        compatible = "maxim,max98357a";
        #sound-dai-cells = <0>;
        pinctrl-names = "default";
        pinctrl-0 = <&pinctrl_sai1>;
        sdmode-gpios = <&gpio1 5 GPIO_ACTIVE_HIGH>;
    };
    // Sound-card definition
    sound {
        compatible = "simple-audio-card";
        status = "okay";
        simple-audio-card,name = "max98357a";
        simple-audio-card,format = "i2s";
        simple-audio-card,bitclock-master = <&dailink_master_cpu>;
        simple-audio-card,frame-master = <&dailink_master_cpu>;
        simple-audio-card,routing =
            "HiFi Playback", "SD_MODE",
            "SD_MODE", "Speaker";
        simple-audio-card,widgets =
            "Speaker", "Speaker";
        simple-audio-card,codec {
            clocks = <&audio_blk_ctrl IMX8MP_CLK_AUDIO_BLK_CTRL_SAI1_MCLK1>;
            sound-dai = <&codec_ext>;
        };
        dailink_master_cpu: simple-audio-card,cpu {
            sound-dai = <&sai1>;
        };
    };
};
&nau8822_1a {
    status = "disabled";
};
/* VERDIN I2S_1 */
&sai1 {
    assigned-clock-parents = <&clk IMX8MP_AUDIO_PLL2_OUT>;
    assigned-clock-rates = <12288000>;
    status = "okay";
};
```

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6 Add Volume Control for MAX98357A

To implement **volume control** for the **MAX98357A I2S Class-D Amplifier**, you can add a **softvol** control by editing the `.asoundrc` file. This method allows you to control the volume of the audio playback without hardware support for a mixer.

6.1 Step 1: Edit the `asoundrc` File

Creating a new **softvol** device

Open the `.asoundrc` file in your home directory with your preferred text editor. If the file does not exist, you can create it:

```
$ vi ~/.asoundrc
```

6.2 Step 2: Add the Softvol Configuration

Add the following configuration to define a new **softvol** control. This creates a virtual mixer that allows volume adjustments for the `max98357a` device:

```
pcm.softvol{
    type      softvol
    slave{
        pcm    "sysdefault:CARD=max98357a"
    }
    control{
        name    "Master"
        card    0
    }
}
pcm.!default{
    type      plug
    slave.pcm "softvol"
}
```

This configuration:

- Defines a new `pcm.softvol` device with a volume control named **Master**.
- Assigns the `max98357a` device as the slave PCM for audio output.
- Sets the **softvol** as the default audio output device (`pcm.!default`).



6.3 Step 3: Apply the Configuration

Once you have added the configuration, save the file and restart the ALSA service to apply the changes:

```
$ systemctl restart alsasound
```

6.4 Step 4: Test the Volume Control

To verify that the **Master** control has been successfully created, use the `amixer` command to list the available mixer controls:

```
root@verdin-imx8mp-14748057:~# amixer
Simple mixer control 'Master',0
  Capabilities: volume
  Playback channels: Front Left - Front Right
  Capture channels: Front Left - Front Right
  Limits: 0 - 255
  Front Left: 255 [100%]
  Front Right: 255 [100%]
```

You can now adjust the volume using the Master control, either through the `amixer` command or `alsamixer`:

```
$ amixer set Master 50%
```



6.5 Step 5: Use alsamixer for Volume Control

\$ alsamixer

```
carriertronic@carriertronic-desktop: ~  
Card: max98357a  
Chip:  
View: F3:[Playback] F4: Capture F5: All  
Item: Master
```

```
lqqk  
xaax  
xaax  
xaax  
xaax  
xaax  
xaax  
xaax  
xaax  
xaax  
xaax  
xaax  
xaax  
xaax  
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xaax  
xaax  
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xaax  
xaax  
xaax  
xaax  
xaax  
xaax  
xaax  
xaax  
xaax  
mqqj  
100<->100  
-<Master>
```

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7 Fixing ALSA Configuration for MAX98357A

In this chapter, we will address how to resolve issues with **ALSA** and **alsamixer** when using the **MAX98357A I2S Class-D Amplifier** with the **Verdin Pico-ITX Dual Ethernet Carrier Board** running the **Linux 5.15.129-6.4.0** build from Toradex.

7.1 Problem Description

After testing the **MAX98357A overlay** on the **Linux 6.4 build** from Toradex, audio playback worked perfectly using **aplay**, but there was an issue with **alsamixer**:

```
root@verdin-imx8mp-14748057:~# alsamixer
ALSA lib ../../alsa-lib-1.2.6.1/src/confmisc.c:165:(snd_config_get_card) Cannot get card index for
imx8mpnau8822
cannot open mixer: No such device
```

This issue arose because the `/etc/asound.conf` file was still configured for the **imx8mpnau8822** codec. To fix this, the codec needs to be replaced with **max98357a**.

7.2 Step 1: Check Linux Version and Audio Device

```
root@verdin-imx8mp-14748057:~# uname -a
Linux verdin-imx8mp-14748057 5.15.129-6.4.0+git.67c3153d20ff #1 SMP PREEMPT Wed Sep 27
12:30:36 UTC 2023 aarch64 aarch64 aarch64 GNU/Linux
```

```
root@verdin-imx8mp-14748057:~# aplay -l
**** List of PLAYBACK Hardware Devices ****
card 0: max98357a [max98357a], device 0: 30c10000.sai-HiFi HiFi-0 [30c10000.sai-HiFi HiFi-0]
Subdevices: 1/1
Subdevice #0: subdevice #0
```



7.3 Step 2: Update the ALSA Configuration

```
$ nano /etc/asound.conf
```

Replace all instances of imx8mpnau8822 with max98357a. Here is an example configuration:

```
defaults.pcm.rate_converter "linear"
pcm.default_card{
    type hw
    card "max98357a"
}
...
```

7.4 Step 3: Test Audio and Mixer

After updating the configuration, reboot the system or restart the ALSA service:

```
$ reboot
# or
$ systemctl restart alsa
```

Test the audio playback again with aplay and verify the functionality of **alsamixer**:

```
$ aplay /path/to/soundfile.wav
$ alsamixer
```




8 Reference documents

Description	Link
Verdin Family Specification	https://docs.toradex.com/109262-verdin-family-specification.pdf
Verdin Computer on Module Family Overview	https://www.toradex.com/de/computer-on-modules/Verdin-arm-family
Verdin Pico-ITX Dual Ethernet Carrier Board	https://carriertronic.com/wp-content/uploads/2024/04/Verdin-Pico-ITX-Dual-Ethernet-Carrier-Board-datasheet.pdf
Verdin Pico-ITX MAX98357A Extention	https://carriertronic.com/4devs/
MAX98357A Class D Amplifier	https://www.analog.com/media/en/technical-documentation/data-sheets/MAX98357A-MAX98357B.pdf



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